



Final Year Project

“Free Space Optical Communication System Using Laser Diode and Arduino”

BS PHYSICS

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CERTIFICATE

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ABSTRACT

Free Space Optical (FSO) communication is an emerging wireless communication technology that uses light instead of radio waves to transmit information through free space. It enables secure and high-speed communication between two nodes and offers advantages, including high bandwidth and immunity to electromagnetic interference. This kind of communication is commonly used by satellites, military communication systems and in industrial automation. The objective of this project was to design and develop a low-cost Free Space Optical Communication System using a laser diode, a BPW34 photodiode, LM358 operational amplifier, and Arduino Uno microcontrollers.

The system transmits data through a laser beam and receives it using a photodiode sensor. The transmitter module uses an Arduino Uno to control a laser module, encoding digital data through on-off keying (OOK) modulation. At the receiver end a sensitive photodiode detects the transmitted optical signal and converts it into a weak electrical signal. This signal is amplified using a LM358 operational amplifier before being processed by the receiver Arduino, which reconstructs and displays the original message on a computer through the serial monitor. In summary, the transmitter converts electrical signals into optical signals using a laser module, and the receiver converts optical signals back into electrical signals using a photodiode.

The developed prototype successfully demonstrated reliable short range optical communication under clear line of sight conditions. However, the communication quality was affected by factors such as misalignment of the transmitter and receiver, ambient light interference, and increasing transmission distance. Furthermore, the proposed system provides a practical introduction to optical wireless communication and offers a foundation for future improvements, including higher data transmission rates, longer communication distances (like between moon and earth), and advanced modulation and error-correction techniques.

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CHAPTER 1

INTRODUCTION

1.1 Problem Statement

Traditional wireless systems face issues such as:

Electromagnetic interference: unwanted signal from other electronic devices

Limited bandwidth: limited amount of space

Spectrum licensing: Government permission to use certain RF

Security vulnerabilities: weakness in communication about security

This project aims to develop a free space optical communication system using laser light to reduce several limitations over specific applications.

1.2 Project Overview

Wireless communication enabling data transfer without physical connections, while radio frequency (RF) communication dominates the wireless landscape. Optical wireless communication using laser technology offers unique advantages including high bandwidth, immunity to electromagnetic interference, and enhanced security.

This project explores the implementation of a simple yet effective laser-based wireless communication system using readily available components. The same communication principle is employed in advanced space optical communication systems (like communication link between moon and earth). In this project Arduino Uno microcontroller serves as the processing unit for both transmitter and receiver modules. Laser module receives digital signals from Arduino and convert them to laser light on transmitter side which place at Earth station and these signals are received by the receiver place at Moon station convert optical signals into digital signals again by photodiode and show the output on screen.

Overall, the project demonstrates the basic principles of optical communication in a practical and efficient way. This system is effective for educational and prototyping purposes.

1.3 Objectives

The primary objectives of this project are:

- Design and implement a laser-based wireless communication system using Arduino Uno
- Develop transmitter and receiver circuits for data transmission
- To learn and implement Arduino programming using the Arduino IDE
- Evaluate the system performance
- Demonstrate practical applications of wireless communication by using laser

1.4 Working Principle

The system works on the principle of light modulation:

- Arduino transmitter sends digital signals
- The laser module converts digital signals into light pulses
- The photodiode detects light pulses by phenomena of photoelectric effect
- Arduino receiver processes and decodes the signals
- Decoded message is displayed on the computer screen using the Arduino Serial Monitor.

CHAPTER 2

DESCRIPTION OF COMPONENTS

2.1 Hardware Components

Table 2.1: Components Used

Sr. No.	Component	Quantity	Description
1	Arduino Uno	2	Microcontroller board for TX and RX
2	Laser Diode Module	1	650nm red laser transmitter
3	Photodiode BPW34	1	For light detection
4	Op-Amp LM358	1	For signal amplification
5	2N2222 Transistor	1	NPN transistor for laser switching
6	Resistors	Various	1 k, 10 k, 220-ohm resistors
7	Breadboard	1	Circuit prototyping board
8	Jumper Wires	Various	Connecting wires
9	Power Adapter	1	Power Supply

2.1.1 Arduino Uno

The Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328 microcontroller and developed by Arduino.cc. The board is equipped with six input pins A0 to A5 that used to read the signal from an analog sensors like temperature sensors and convert it into digital values that read by microcontroller. The board has 14 digital I/O pins as shown in Figure 2.1, of which six can be used as PWM (Pulse Width Modulation) output, 6 analogs inputs pins, and is programmable with the Arduino IDE (Integrated Development Environment) via a type B USB cable. It also has 16 MHz quartz crystal and this board can be powered by the USB cable or by an external 9-volt battery, though it accepts voltages between 7 to 12volts.

The word "Uno" means "one" in Italian and was chosen to mark the initial release of Arduino Software. The ATmega328 on the board comes preprogrammed with a bootloader that allows uploading new code to it without the use of an external hardware programmer.

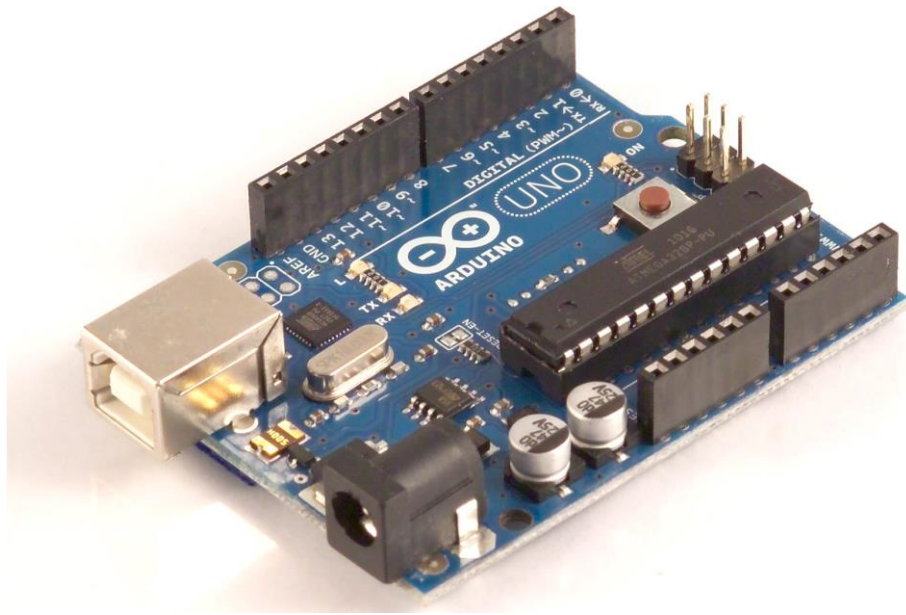


Figure 2.1: Arduino Uno

- **Technical Specifications**

Microcontroller	Microchip ATmega328
Operating Voltage	5Volts
Input Voltage	7 to 12Volts
Digital I/O Pins	14 (of which 6 provide PWM output)
Analog Input Pins	6
DC Current per I/O pin	20 mA
DC Current for 3.3V pin	50 mA
Flash Memory	32 KB, 0.5 KB used by bootloader
SRAM	2 KB
EEPROM	1 KB
Clock Speed	16 MHz
Length	68.6 mm
Width	53.4 mm

- **Microcontroller (ATmega328P)**

The most powerful part of this board is Microcontroller, which is known as brain of this. The high-performance Microchip Pico Power 8-bit AVR RISC-based microcontroller combines 32KB ISP flash memory with read-while-write capabilities.

The microcontroller consists twenty-eight pins including analog pins, digital pins, and power

pins, as shown in Figure 2.2. This microcontroller controls all the functionalities and working of Arduino Uno and there attached components.

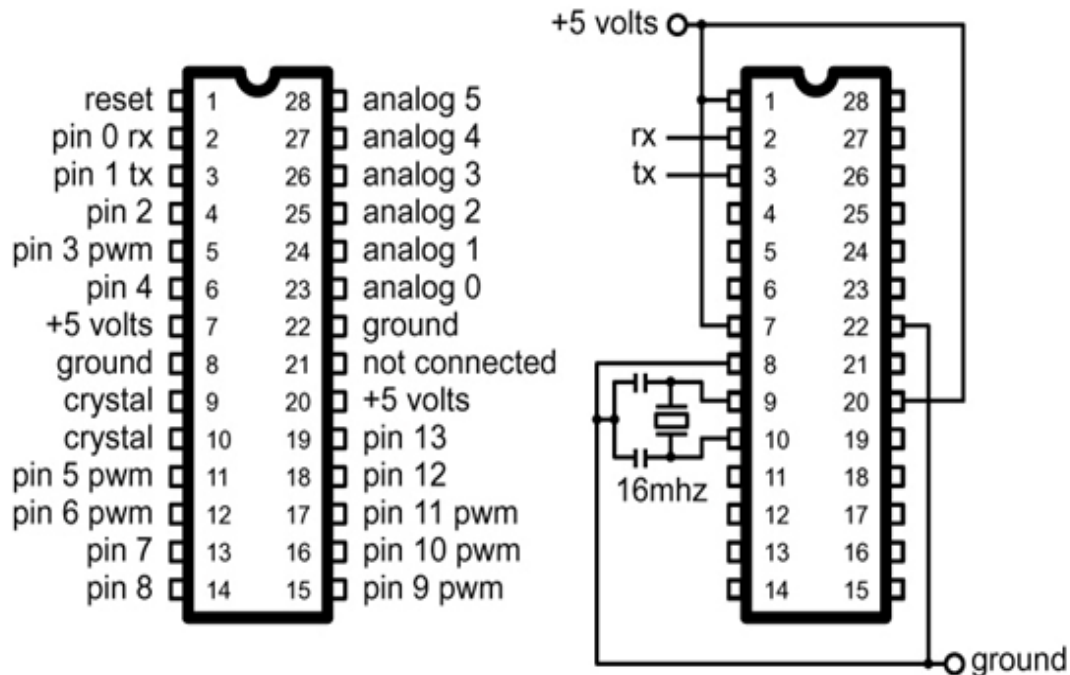


Figure 2.2: Arduino Uno Microcontroller pin configuration

2.1.2 Laser Module

The 5 V laser module is a key component used in Free Space Optical (FSO) communication systems for transmitting information in the form of a **coherent light beam**. LASER stands for *Light Amplification by Stimulated Emission of Radiation*. This module emits a **highly focused**, **monochromatic**, and **directional beam** of light when supplied with a 5 V power source.

In this project, the laser module functions as the **transmitter**. It converts electrical signals into corresponding optical signals, which are then propagated through free space to the receiver. The intensity of the laser beam is modulated in accordance with the input signals which enabling the transmission of information without the need for physical cables or radio frequency communication.

The 5 V laser module generally includes a laser diode along with a **built-in driver** circuit. That ensure stable operation and protection of the diode from fluctuations in current. It typically consists of three terminals, as shown in Figure 2.3:

1. VCC (power supply)
2. GND (ground)
3. Signal (input control)

The signal terminal is interfaced with the Arduino or signal source, which controls the modulation of the laser output.

Due to its high directionality and low beam divergence, the laser module is capable of transmitting signals over relatively long distances with minimal dispersion. This makes it suitable for point-to-point communication systems.

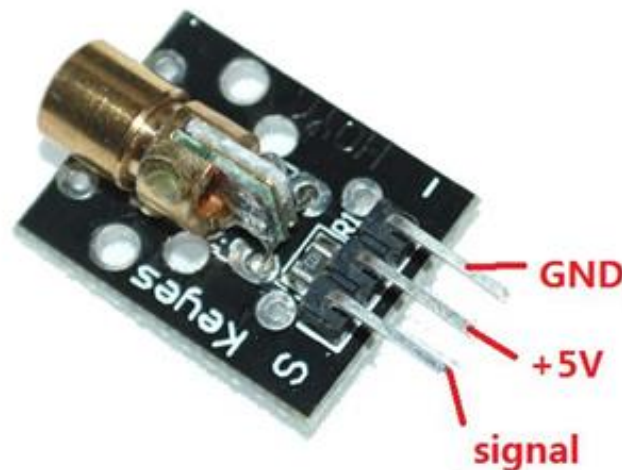


Figure 2.3: Arduino Laser Diode Module

- **Characteristics Of Laser Module ($T_c=25^\circ\text{C}$)**

Threshold current:	15 – 30 mA
Operating current:	20 – 40 mA
Operating voltage:	2.3 – 2.8 V
Output efficiency:	0.85 - 1.20 W/A
Monitor current:	0.30 - 0.60 mA
Lasing wavelength:	655 – 670 nm

2.1.3 Photodiode (BPW34)

The BPW34 is a silicon **PIN photodiode** designed for the detection of optical radiation and its conversion into an electrical signal. This operates on the principle of the **photoelectric effect**, in which incident photons generate electron-hole pairs within the depletion region of the semiconductor, and flow current name as photoelectric current which is proportional to intensity of incident light. Due to its high sensitivity, fast response time, and low noise characteristics, the BPW34 is widely employed in optical communication and sensing applications. This photodiode has two terminals, as shown in Figure 2.4, one is Anode and other is Cathode and cathode terminal can determine by cathode marking on that side.

In this project, the BPW34 photodiode serves as the **receiver element**. It detects the **modulated laser beam** transmitted from the laser module and converts the optical signal into an equivalent electrical signal. The variations in the intensity of the incident light are translated into corresponding variations in photocurrent, which are then further processed, amplified, and reconstructed to retrieve the original transmitted information, such as text or audio signals.

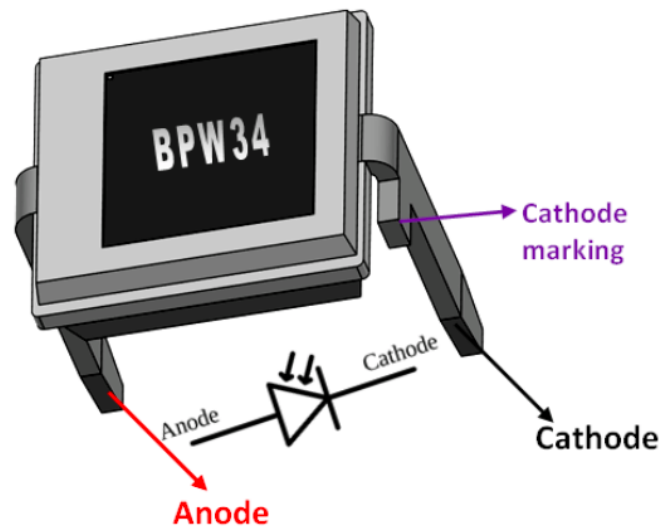


Figure 2.4: Photodiode BPW34

The device exhibits fast switching characteristics that making it suitable for detecting high-frequency modulated optical signals used in communication systems. Besides it offers good spectral response in the visible and near-infrared regions, which aligns well with the emission characteristics of common laser sources used in such projects.

- **Characteristics of BPW34**

Breakdown voltage (V_{BR}):	Min. 60 V
Reverse dark current (I_{ro}):	2 nA – 30 nA
Open circuit voltage (V_o):	350 mV
Short circuit current (I_k):	70 μ A
Reverse light current (I_{ra}):	50 μ A – 75 μ A
Maximum Power dissipation (P_V):	215 mW
Wavelength of peak sensitivity (λ_p):	900 nm
Rise and Fall time (t_r & t_f) at $V_R = 10$:	100 ns

2.1.4 LM358 Operational Amplifier (Op-Amp)

The LM358 is a dual operational amplifier integrated circuit that consists of two independent, high-gain, internally frequency-compensated. Op-amps designed to operate from a single

power supply. It is widely used in signal conditioning, amplification, filtering, and processing applications due to its low power consumption, wide supply voltage range, and ease of use. In this project, the LM358 op-amp is used to amplify the weak electrical signal obtained from the photodiode (BPW34). Since the output of the photodiode is typically very small in magnitude, it requires amplification before it can be processed further. The LM358 helps in boosting the signal to a usable level while maintaining the integrity of the original information.

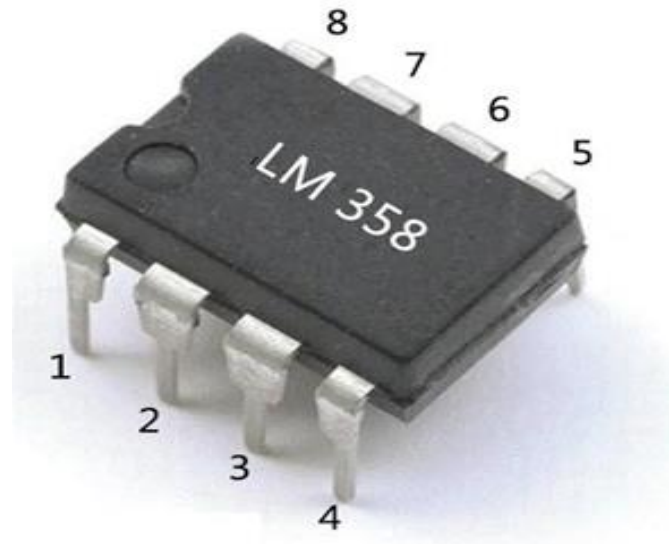


Figure 2.5: Op-Amp LM 358

Here pin configuration of Op-Amp is:

Pin1 and **pin7** are output

Pin2 and **pin6** are inverting input

Pin3 and **pin5** are non-inverting inputs

Pin4 is the GND terminal

Pin8 is VCC+ positive supply

- **Characteristics of Op-Amp LM358**

Max. Power Supply Voltages (V_{cc}):	32 V _{dc}
Input Offset Voltage (V_{IO}):	2 mV – 3 mV
Input Offset Current (I_{IO}):	5nA – 30nA
Slew Rate:	0.5 V/ μ s
Common Mode Rejection (C_{MR}):	70 dB
Open Loop Voltage Gain (A_{VOL}):	100 V/mV
Output Voltage (V_o):	5.0 mV – 3.5 V

Pins diagram and internal structure are shown in Figure 2.5 and in Figure 2.6, in details

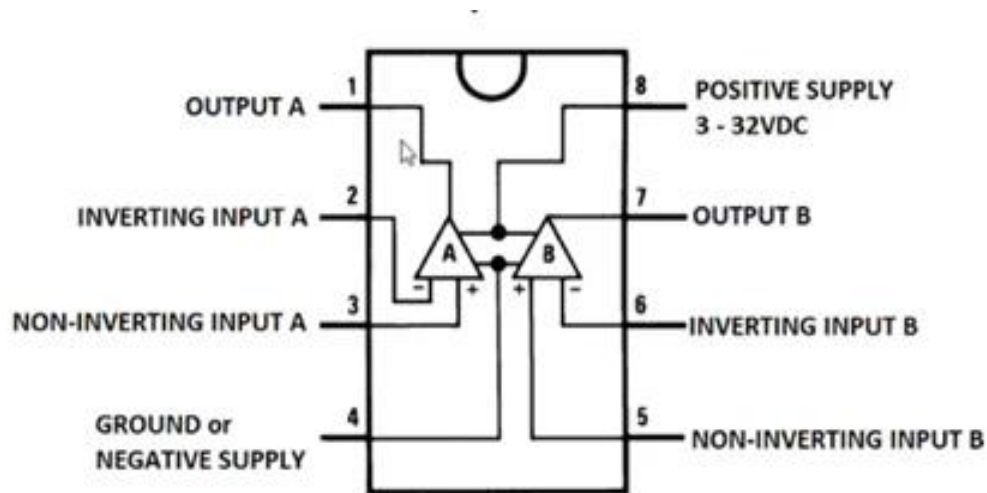


Figure 2.6: Op-Amp LM 358 Top View

2.1.5 NPN Transistor

An NPN transistor is a type of bipolar junction transistor (BJT) that is widely used for amplification and switching applications in electronic circuits. It consists of three semiconductor layers arranged in the form of **N-type**, **P-type**, and **N-type** materials, forming two PN junctions. The three terminals of an NPN transistor are:

1. **Emitter**
2. **Base**
3. **Collector**

as shown in Figure 2.7.

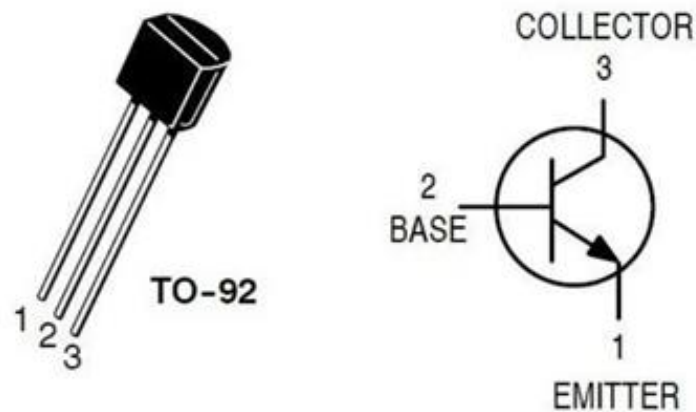


Figure 2.7: NPN Transistor

Collector is large and base is small terminal of the transistor. Usually for amplify the signals emitter kept ground or say common. And for biasing the transistor emitter-base junction will forward bias and collector-base junction should be reverse bias.

In an NPN transistor, a small current applied to the base terminal which controls a larger current flowing from the collector to the emitter. This property allows the transistor to act as a current amplifier as well as an electronic switch.

In this project, the NPN transistor is used as **control switch** required for driving the laser module. Since the output current from the Arduino is limited, the transistor acts as a driver to provide sufficient current to operate the laser diode effectively. It ensures that the laser module receives a stable and controlled current, preventing damage to the components.

- **Characteristics Of NPN**

Collector–Emitter Breakdown Voltage ($V_{(BR)CEO}$):	40 V _{dc}
Collector–Base Breakdown Voltage ($V_{(BR)CBO}$):	75 V _{dc}
Emitter–Base Breakdown Voltage ($V_{(BR)EBO}$):	6.0 V _{dc}
Collector Cutoff Current (I_{CEX}):	10 nAdc
Collector Cutoff Current (I_{CBO}):	10 μ Adc
Emitter Cutoff Current (I_{EBO}):	10 nAdc
Base Cutoff Current (I_{BEX}):	20 nAdc
DC Current Gain (h_{FE}):	100–300
Output Capacitance (C_{ibo}):	8 pF
Input Impedance (h_{ie}):	2–8 K
Small Signal Current Gain (h_{fe}):	75–375
Output Admittance (h_{oe}):	25–200 μ Mhos

2.1.6 Resistor

Resistor is a **passive electronic component** that provide a fixed resistance say 1 K and 220ohms in an electrical circuit. It is used to control the flow of electric current and to provide voltage drops at specific points in a circuit according to Ohm's Law. Resistors are essential components in almost all electronic systems, as they help in protecting sensitive devices and ensuring proper circuit operation.

In this project, the resistors are used primarily for **current limiting** and **biasing purposes**. It helps regulate the amount of current flowing through components. Here resistor also plays a role in maintaining signal integrity within the circuit. It can be used in conjunction with the

photodiode or other components to help stabilize the signal and reduce fluctuations caused by noise or variations in input signals.



Figure 2.8: Resistor

Resistors are typically color-coded to indicate their resistance value, tolerance, and sometimes power rating.

2.1.7 Breadboard

A breadboard as shown in Figure 2.9, is a reusable **prototyping tool** used to build and test electronic circuits without soldering. It consists of a plastic board with a grid of interconnected metal clips beneath the surface, which allow electronic components and wires to be inserted and connected easily. The internal connections are arranged in a systematic way.

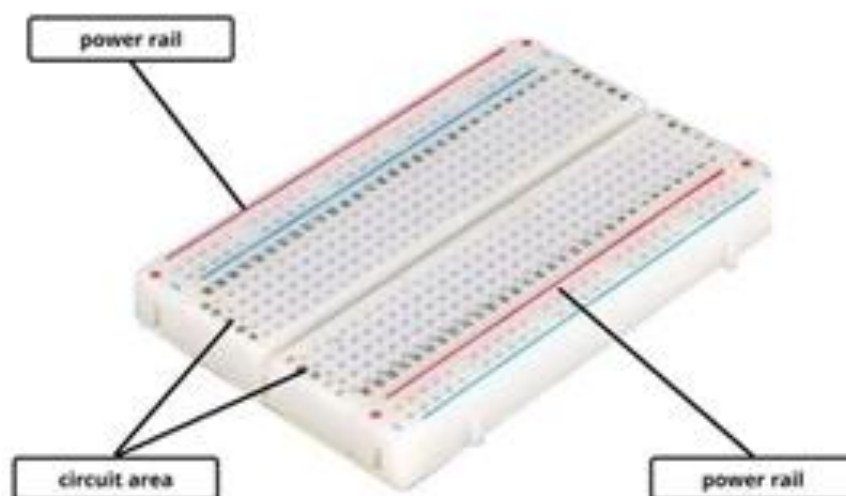


Figure 2.9: Breadboard

Here, the breadboard is used to assemble and interconnect all the components, such as the Arduino, laser module, NPN transistor, photodiode, and resistors. It provides a convenient platform for testing and modifying the circuit during the development phase without making permanent connections.

The power rails on the sides of the breadboard are commonly used to distribute **VCC (power supply)** and **GND (ground)** across the circuit, ensuring a common reference point for all components. The **central area** is used for placing integrated circuits and other discrete components, allowing proper connectivity between them.

2.1.8 Jumper Wires

Jumper wires are simply wires that have connector pins at each end, allowing them to be used to connect two points to each other without soldering. Jumper wires are typically used with breadboard and other prototyping tools to make it easy to change a circuit as needed.

Jumper wires typically come in three versions:

- **Male-to-male**
- **Male-to-female**
- **Female-to-female**

The difference between each is in the end point of the wire. Male ends have a pin protruding and can plug into things, while female ends do not pin and are used to plug things into it. Male-to-male jumper wires are the most common and likely use in prototyping and connecting two ports on a breadboard.

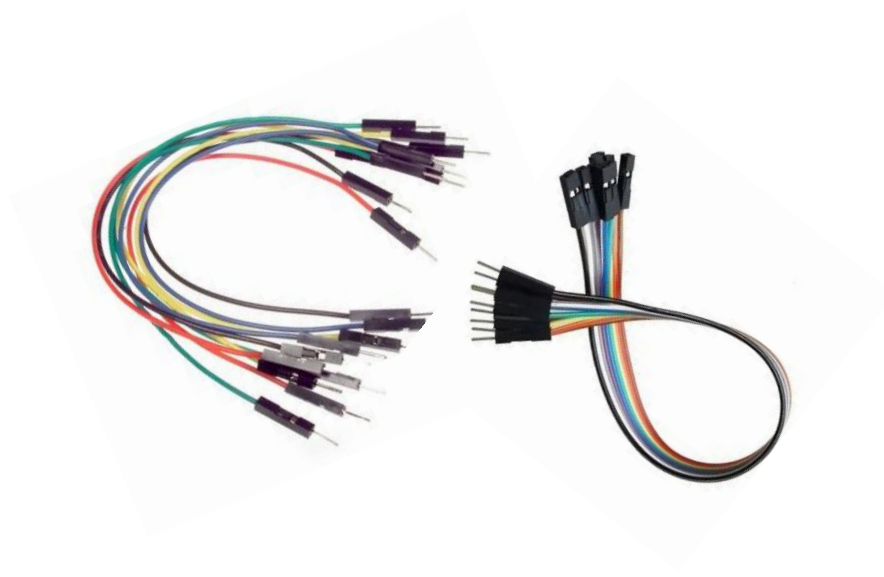


Figure 2.10: Jumper Wires

2.1.9 Power Adapter

Power Adapter is a type of **external power supply**, often enclosed in a case like to an AC plug. AC adapters deliver electric power to devices that lack internal components to draw voltage and power from mains power themselves. The internal circuitry of an external power supply is often very similar to the design that would be used for a built-in or internal supply. Here power adapter is used **about 5volts and 2amperes**, which power the whole circuit and peripherals of Arduino Uno.



Figure 2.11: Power Adapter

CHAPTER 3

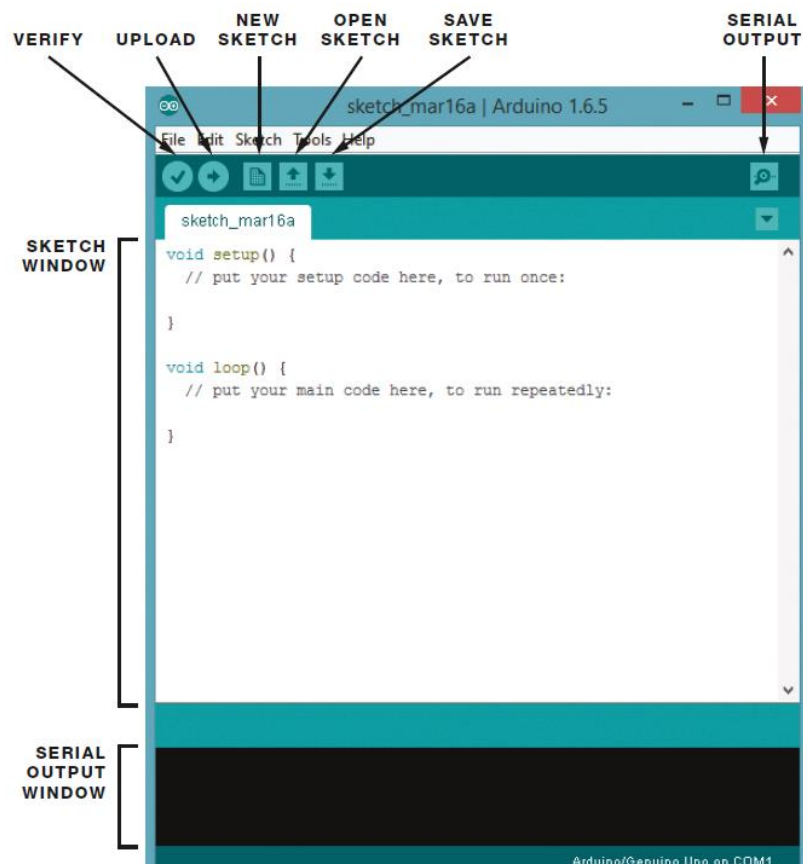
DESIGN AND CIRCUIT

3.1 Software

3.1.1 Arduino IDE

The Arduino Integrated Development Environment (IDE) is an open source and official software design by Arduino.cc. It is cross-platform application (for Windows, macOS, Linux) that is written in functions from **C** and **C++**. It is used to write and upload programs to Arduino compatible boards, but also with help of 3rd party cores, other vendor development boards.

Programs written using Arduino Software (IDE) are called **sketches**. These sketches are written in the text in the text editor and are saved with the file extension **.ino**. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and displays error. The console displays text output by the Arduino Software (IDE), including complete error messages and configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor.



- **Arduino Libraries**

In the Arduino world, a library is a small piece of code that carries out a specific function. Rather than enter this same code repeatedly in your sketches, you can add a command that borrows code from the library. This shortcut saves time and makes it easy for you to connect to items such as a sensor, display, or module. The Arduino IDE includes several built-in libraries, such as the Liquid Crystal library, which makes it easy to talk to LCD displays and there are many more available online. we can find all of the libraries which we need at <http://www.nostarch.com/arduinohandbook/>.

3.2 Connections

Connections of both Transmitter and Receiver slides.

3.2.1 Transmitter Side

The Laser Module

- Connect the **Positive (VCC)** wire of the 5 V laser Module directly to the **5 V Power Rail**.
- Connect the **Negative (GND/Signal)** wire of the laser Module to the **Collector (pin 1)** of the transistor.

The Base Control (Voltage Divider):

- Connect the **10 k Ω resistor** from the Arduino data pin to the **Base (pin 2)** of the transistor.
- Connect the **1 k Ω resistor** from the **Base (pin 2)** to **GND**. (*This ensures that when the Arduino outputs 5 V, the voltage at the base is tightly regulated*).

The Current Setter (Emitter Side):

- Connect a low-value resistor (typically between 10 Ω to 47 Ω depending on laser's current rating) from the **Emitter (pin 3)** to **GND**.

3.2.2 Receiver Side

- **Photodiode (BPW34):** Connect the **Cathode** to **5 V**. Connect the **Anode** to one end of the **10 k Ω resistor**. Connect the other end of the 10 k Ω resistor to **GND**.
- **Op-Amp Inputs:** Connect a jumper wire from the junction between the photodiode anode and the 10 k Ω resistor to **LM358 pin 3 (+)**.
- **Gain Feedback:** Place the **100 k Ω resistor** between **LM358 pin 1 (Output)** and **pin 2**.
- **Gain Grounding:** Place the **1 k Ω resistor** between **LM358 pin 2 (-)** and **GND**.
- **Decoupling:** Place the **0.1 μ F capacitor** directly between **LM358 pin 8 (5 V)** and **pin 4**.
- **To Arduino:** Connect **LM358 pin 1 (Output)** to Arduino **Digital pin 2** (we will use a digital reading with a software serial implementation for robust edge detection).

3.3 Working

Working of both, transmitter and receiver side is as follow:

3.3.1 Transmitter Side

In transmitter side the main component is laser module. This laser module is attached with Arduino Uno and the Arduino Uno is attached to serial monitor say or displayed on a computer screen. When any text or message is typed on screen and give the command to Arduino Uno, this sent the message to laser module in the form of digital signals or say in the form of binary data, then the laser module converts this digital signal into optical signal and transmit it.

Between the laser module and Arduino here NPN transistor and a base resistor also use, the task of these components to protect Arduino pins, limits current into transistor base and to keep the transistor in saturation mode.

- **Encoding and Transmission**

Here when the data is given to Arduino, it converts each data or each letter into an 8-bit binary code and this code flash the laser according to pattern. When there is binary '1' in code laser intensity switched ON and when there is binary '0' laser intensity Off, this is known as **Modulation On-Off Keying (OOK)**.

- **Modulation**

“Modulation is a process of changing the characteristics of the wave to be transmitted of the wave to be transmitted by superimposing the message signal on the high frequency signal”.

These modulation techniques are classified into two major types:

Analog Modulation and **Digital Modulation**.

In **Analog Modulation** a continuously varying sine wave is used as a carrier wave that modulates the data signals. And in **Digital Modulation** the carrier wave is Keyed or switched On and Off to create pulses such that the signal is modulated.

Here, Digital Modulation is used in laser module in the form of **On-Off keying Modulation** which simply mean, laser will be ON when binary data is '1' and laser will be OFF when binary data is '0'.

Mathematical Expression:

$$s(t) = \begin{cases} A \cos(2\pi ft) & \text{for binary 1} \\ 0 & \text{for binary 0} \end{cases} \quad (3.1)$$

Here:

A = Amplitude

f = frequency

t = time

$s(t)$ = transmitted signal

- **Components Use**

1. Arduino Uno
2. Laser module (5 volt)
3. NPN Transistor (BC547)
4. 1 K Ω Resistor
5. Breadboard
6. Jumper wires

3.3.2 Receiver Side

In receiver side there is **PIN Photodiode BPW34** is important and special component. PIN Photodiode receive the modulated optical signals or say light pulses from transmitter (laser module) and when light fall on it there generate electron-hole pair, in reverse bias electric field pulls charges which produce electric current, that known as **Photo-electric Current**.

This photoelectric current is much weak which is amplified by **Op-Amp LM358**. This operational amplifier is specially designed to operate from a single power supply over a wide range of voltage, and it is more flexible for lower voltage.

- **Decoding and Display**

When Op-Amp boosts the received signal from photodiode than this was send to pin 8 of Arduino Uno, then Arduino read the sequence of signal and decode it again in 8-bit binary code. This binary code is then converted into original message by Arduino and then this display on screen.

- **Components Use**

1. Arduino Uno
2. Photodiode (BPW34)
3. Op-Amp LM358
4. Resistor (10 k Ω , 100 k Ω)
5. Breadboard
6. Jumper wires
7. Display screen

3.3.3 Working Diagram

This is the complete sketch of this project which are shown in Figure 3.1. Here step by step show that how data transmit through transmitter sequentially and then receive by receiver. In this diagram amplifier also show which amplify the receiving signal and then Arduino which decode and display the message.

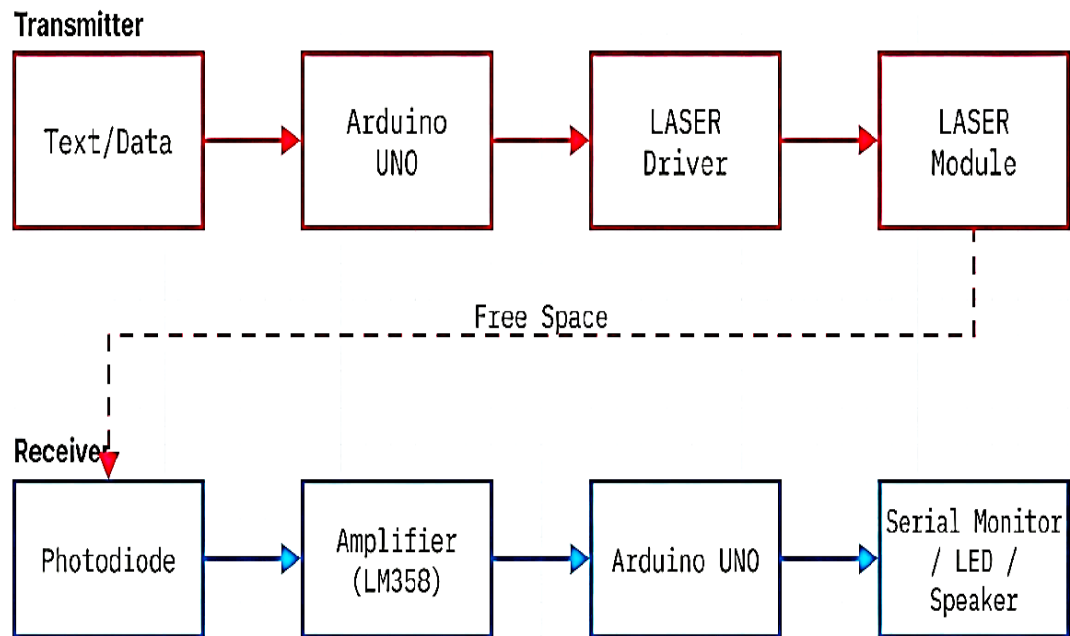


Figure 3.1: Complete Project Working Concept

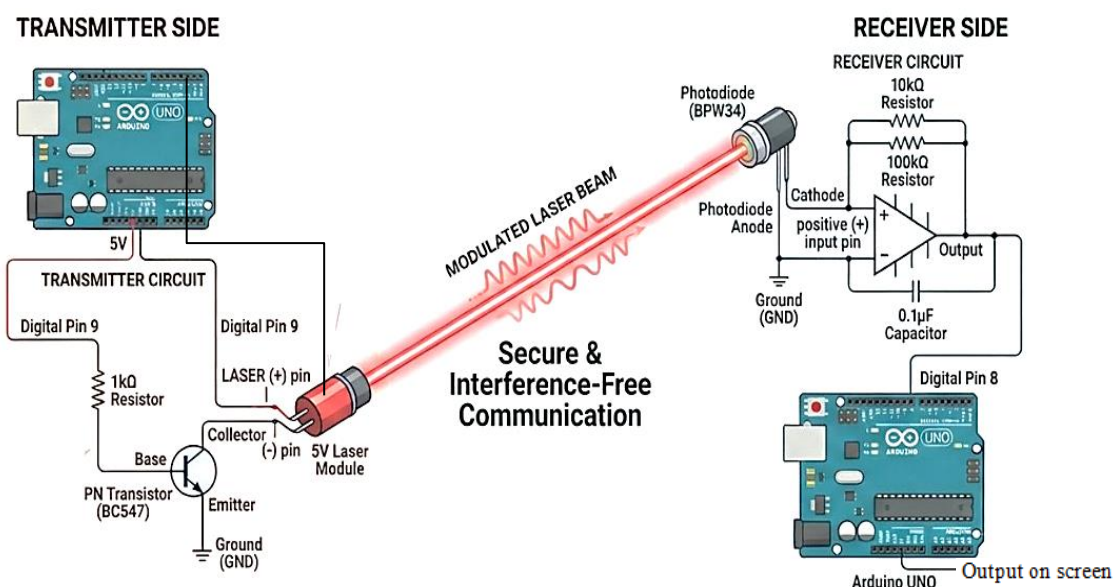


Figure 3.2: Complete Project Circuits Diagrams

3.3.4 Program Code

The program code for transmitter and receiver side is as follow:

- **Transmitter Code**

```
#include <SoftwareSerial.h>

// pin 3 is TX (connect to laser), pin 2 is RX (unused here)
SoftwareSerial laserSerial(2, 3);

void setup() {
  Serial.begin(9600); // Hardware serial for Computer
  laserSerial.begin(1200); // Low baud rate for higher reliability over the air
  Serial.println("Laser Transmitter Ready. Type your message below:");
}

void loop() {
  if (Serial.available() > 0) {
    char inChar = Serial.read();
    laserSerial.write(inChar); // Send character via laser
    Serial.print(inChar); // Echo back to local monitor
  }
}
```

- **Receiver Code**

```
#include <SoftwareSerial.h>

// pin 2 is RX (connect to LM358 pin 1), pin 3 is TX (unused here)
SoftwareSerial laserSerial(2, 3);

void setup() {
  Serial.begin(9600); // Hardware serial for Computer
  laserSerial.begin(1200); // Must match the transmitter's baud rate

  Serial.println("Laser Receiver Ready. Waiting for signal...");
}
```

```
void loop() {  
  if (laserSerial.available() > 0) {  
    char outChar = laserSerial.read();  
    Serial.print(outChar); // Display received character on monitor  
  }  
}
```

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Graphical Analysis

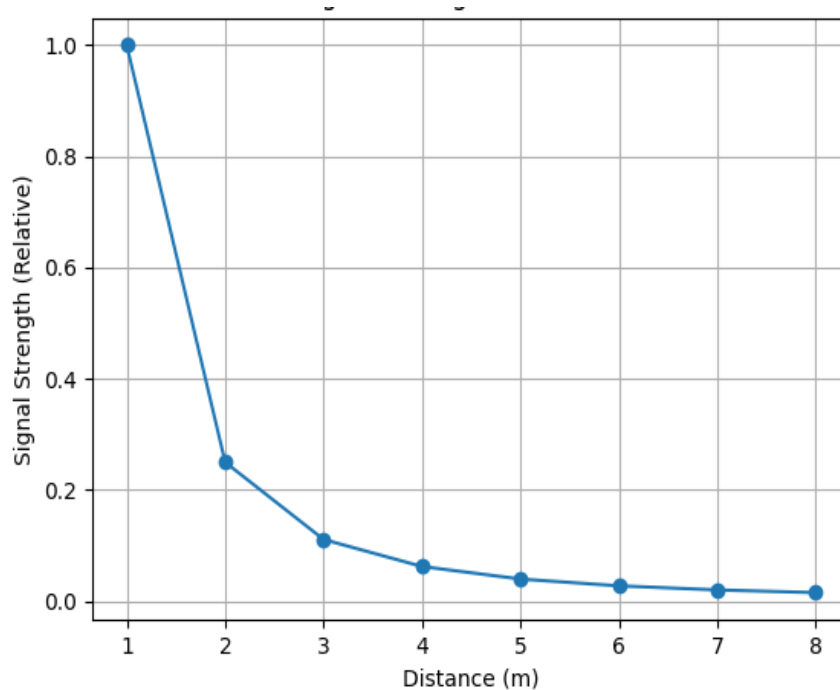
- **Distance vs Signal strength of Laser**

The relation between distance(d) and signal strength (I) is inversely proportional. It mean when we kept receiver away from transmitter out of there minimum range than the signal strength will lose.

Mathematically:

$$I \propto \frac{1}{d^2}$$

So, the graph will be:



- **Light Intensity vs Photodiode Output**

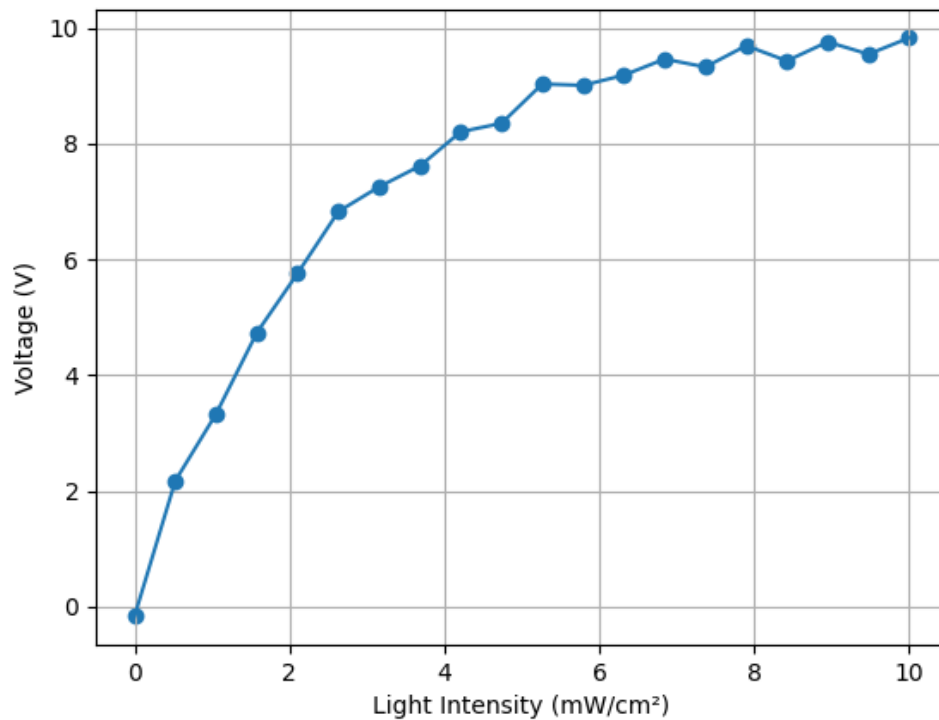
As light intensity will high the output of photodiode will also high in the form of voltage and current. So, **Mathematical relationship** will be:

$$I_{photo} = R \cdot P_{light} \quad (4.1)$$

Here:

$$P_{light} = \text{light power} \quad \text{and} \quad I_{photo} = \text{current generated}$$

And graph will be:



4.2 Transmission Distance Analysis

The proper communication between transmitter and receiver is bounded in specific distance between two channels. And that distance is:

Distance	Performance
1-2 m	Excellent
2-3 m	Good(stable)
3-7 m	Moderate (some noise)
> 7 m	Degraded

These results were obtained experimentally under indoor laboratory conditions using a [650 nm](#) laser module.

4.3 Effect of Noise

Sunlight and artificial light introduce some noise which distorted the signals and can disturb the proper communication. So, for this Optical shielding and filtering capacitor are used, which reduce the noise effect.

4.4 End-to-End Communication Results

Different messages were transmitted, say “HELLO”, “12345678”, “Laser”.

And all these messages were received correctly, according to transmitter program.

CHAPTER 5

CONCLUSION AND FUTURE EXPANSION

5.1 Conclusion

The project free space optical communication system using laser module and Arduino is successfully demonstrate free space communication which offer advantages over wire and radio wave communication. The objective of this project was to introduce a communication system which is independent to any medium for communication and which is fast and most important it will highly directional and secure. So, these things make it more valuable especially for Defense and satellite communication systems, beside this project idea have high potential for futurist advancement with modern technologies

5.2 Applications and Future Expansion

This project has many applications and ability to expand in future on a wide range. Some of the applications are following:

1. Moon to Earth/Earth to Moon Communication

Between moon and earth. signals and communications exchanged by laser beam. This is the future technology and these systems are under development. NASA deep-space mission is working on this project to make such laser which can travel between moon and earth and carry signals without any distortion.

2. Satellite and Space Communication

In space, satellites use laser communication instead of radio waves communication due to its very high data rate and less interference.

3. Military and Secure Communication

In defense and military system, laser communication is widely use due to high security and it is very hard to intercept. Besides, narrow laser beam makes it more secure and difficult to hack.

4. Li-Fi Technology

Li-fi technology some kind of future internet which use light instead of Wi-Fi. It is much faster than traditional internet mode.

5. Industrial Automation

laser communication use in industrial unit for sensors and machine-to- machine communication. For error detections laser signals are also use. Besides, it is reliable and safe for industries.

6. Medical Technology

In medical field optical signals used in imaging, diagnosis, and data transmission between medical devices.

5.3 Limitations

1. laser communication requires clear line of sight.
2. It is affected by fog, rain and dust.
3. Their speed is limited by Arduino timing delay.
4. Require precise alignment between transmitter and receiver.

References

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